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NPTEL (<https://swayam.gov.in/explorer?ncCode=NPTEL>) » Introduction to Graph Algorithms (course)

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Course
outline

About
NPTEL ()

How does an
NPTEL
online
course
work? ()

Week 1
Introduction
and
Principles of
Algorithms ()

☐ Introduction
and Principles
of Algorithms
(unit?
unit=18&lesso
n=36)

☐ Principles of
Algorithms
(Part 2) (unit?)

Week1 Assignment 1

The due date for submitting this assignment has passed.

Due on 2025-08-06, 23:59 IST.

Assignment submitted on 2025-08-06, 23:56 IST

1) State True or False

0.25 points

$$2^{n+1} = \theta(2^n)$$

☒ True

☐ False

Yes, the answer is correct.

Score: 0.25

Accepted Answers:

True

2) State True or False

0.25 points

$$2^n = \theta(3^n)$$

☐ True

☒ False

Yes, the answer is correct.

Score: 0.25

Accepted Answers:

False

3) State True or False

0.25 points

$$2^n = \theta(n!)$$

☐ True

☒ False

unit=18&lesso
n=34)

☒ Introduction to
Graphs (unit?
unit=18&lesso
n=35)

☐ Undirected
Graph (unit?
unit=18&lesso
n=37)

☐ Week 1
Lecture
Material (unit?
unit=18&lesso
n=50)

☒ **Quiz: Week1
Assignment 1
(assessment?
name=97)**

☐ Feedback
Form (unit?
unit=18&lesso
n=26)

**Week 2
Shortest
Path
Algorithms
and
problems ()**

**Week 3
Bellman
Equations ()**

**Week 4
Bellman
Ford and
Dijkstra
Algorithm ()**

**Week 5 All
Pair Shortest
Path ()**

**Week 6
Prims and
Kruskal's
Algorithm ()**

Yes, the answer is correct.

Score: 0.25

Accepted Answers:

False

4) State True or False
 $2^n = O(n!)$

☒ True
☐ False

Yes, the answer is correct.

Score: 0.25

Accepted Answers:

True

5) State True or False
 $n^{\frac{1}{2}} + n^{\frac{1}{4}} = \theta\left(n^{\frac{3}{4}}\right)$

☒ True
☐ False

No, the answer is incorrect.

Score: 0

Accepted Answers:

False

6) State True or False
 $n^{\frac{1}{2}} \bullet n^{\frac{1}{4}} = \theta\left(n^{\frac{3}{4}}\right)$

☒ True
☐ False

Yes, the answer is correct.

Score: 0.25

Accepted Answers:

True

7) State True or False
 $1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} = \theta(\log_2 n)$

☒ True
☐ False

Yes, the answer is correct.

Score: 0.25

Accepted Answers:

True

8) State True or False
 $\log_{10} n^2 = \theta(\log_2 n^3)$

☒ True
☐ False

0.25 poi

0.25 points

0.25 points

0.25 points

0.25 points

**Week 7 DFS
and Cut
Vertex ()**

**Video
download ()**

**Live session
()**

Yes, the answer is correct.

Score: 0.25

Accepted Answers:

True

9) Let A and B be two matrices with dimensions $(p \times q)$ and $(q \times r)$ respectively. Let $C = A \cdot B$. What is the dimension of the matrix C ? **0.5 points**

☒ $p \times r$

☐ p/r

☐ $p + r$

☐ $p - r$

Yes, the answer is correct.

Score: 0.5

Accepted Answers:

$p \times r$

10) Let A and B be two matrices with dimensions $(p \times q)$ and $(q \times r)$ respectively. Let $C = A \cdot B$. How many multiplications are performed for computing C ? **0.5 points**

☒ pqr

☐ $p + r + r$

☐ pq/r

☐ p/qr

Yes, the answer is correct.

Score: 0.5

Accepted Answers:

pqr

11) Let A and B be two matrices with dimensions $(p \times q)$ and $(q \times r)$ respectively. Let $C = A \cdot B$. How many additions are performed for computing C ? **1 point**

☒ $pr(q-1)$

☐ $pr(q+1)$

☐ $pq(r-1)$

☐ $pr(q+1)$

Yes, the answer is correct.

Score: 1

Accepted Answers:

$pr(q-1)$

12) Let G be a directed graph with 15 vertices. Assume that G has no self-loops. The maximum number of one's that may be in the Adjacency Matrix of G is ____.

210

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 210

1 point

13) Let G be a directed graph with 12 vertices. Assume that it may have self loops but no anti parallel edge. That is if $(x,y) \in E$, then $(y,x) \notin E$. The maximum number of one's in the Adjacency Matrix of G is ____.

Yes, the answer is correct.

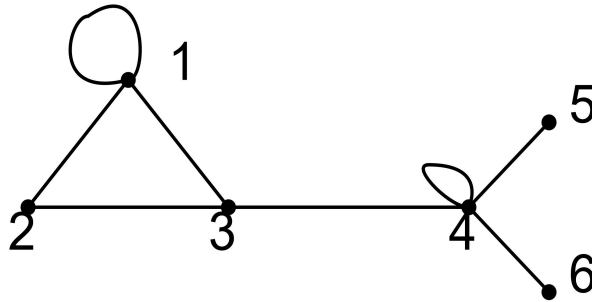
Score: 2

Accepted Answers:

(Type: Numeric) 78

2 points

14) The number of zeroes in the adjacency matrix of the G given below is



Yes, the answer is correct.

Score: 2

Accepted Answers:

(Type: Numeric) 22.0

2 points

15) Let G be an undirected graph with n vertices and no self loops. The maximum number of edges that G may have is

1 point

- ☒ $n(n-1)/2$
- ☐ $n(n+1)/2$
- ☐ $n(n-1)$
- ☐ $n(n+1)$

Yes, the answer is correct.

Score: 1

Accepted Answers:

$n(n-1)/2$